

DE 424 Mini Project - Find the Wumpus

A Probabilistic Graphical Model for Wumpus Tracking

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Declaration

I, Kehan Taljaard, declare that this report and the work presented in it are my own work, except where explicitly acknowledged below. I have not received or accepted information from any source not properly acknowledged in this declaration or elsewhere in the report.

Use of AI assistance. [FILL IN: specific account of AI tool use.]

Other sources. The DE424 lecture notes (Weeks 1–10), the *emdw* user guide and the example code provided with the project, and the textbook by Barber [1] (in particular Chapter 23 on dynamic models) were consulted for theoretical background and library API usage.

ECSA Graduate Attributes

GA 1 – Problem solving. The wumpus tracking problem is formulated from first principles as a hidden Markov model with a uniform-random-walk transition and conditionally independent per-cell Bernoulli observations (Section 2). The model structure is motivated by the problem’s Markov dynamics and the conditional independence of the detection grid given the unobserved position; references to the literature underpinning this formulation are given throughout [1, 2]. A key simplification, collapsing $(T+1)G$ binary detection RVs into $T+1$ unary evidence factors on the unobserved variables, is derived from first principles in Section 2.5. Design trade-offs (exact inference on a chain, point-estimate parameter learning by EM rather than full Bayesian inference, scoping to datasets 1–3) are analysed in Sections 2.8 and 4.

GA 2 – Application of scientific and engineering knowledge. The solution applies probability theory (joint factorisation, marginalisation, conditional independence), graphical-model theory (Bayes networks, factor graphs, cluster graphs), and inference algorithms (sum-product belief propagation, the Expectation–Maximisation algorithm) to the engineering problem of tracking from inaccurate measurements. The C++ implementation using the *emdw* library (Section 3) operationalises the design. Convergence behaviour and numerical considerations are reported in Section 4.

1 Problem description

A wumpus performs a random walk on a $W \times H$ grid over $T+1$ discrete time steps. Its position is unobserved. At each time step, every cell of the grid emits a binary detection: the wumpus’s cell emits a 1 with probability p_w and a 0 otherwise, while every other cell emits a 1 with probability p_c and a 0 otherwise. The entire detection grid is observed at every time step. The task is to infer the wumpus’s trajectory from the sequence of detection grids.

Five datasets are provided. This work addresses datasets 1, 2 and 3. The scoping decision is motivated in Section 2.8.

2 Solution design

2.1 Notation

Symbol	Meaning
W, H	grid width and height (columns and rows)
$G = W \cdot H$	total number of cells
T	final time step index ($T+1$ steps in total)
$\mathcal{X}$	state space, $\{0, 1, \dots, G-1\}$
$X_t \in \mathcal{X}$	unobserved wumpus cell index at time t
$D_t^{(i)} \in \{0, 1\}$	detection RV at cell i at time t
$d_t^{(i)} \in \{0, 1\}$	observed value of $D_t^{(i)}$
$\mathbf{d}_t$	full detection grid at time t
p_w, p_c	detection probabilities (wumpus cell / other cells)
$\gamma_t(x)$	smoothed posterior $P(X_t = x \mid \mathbf{d}_{0:T})$

2.2 Random variables and Bayes network structure

The unobserved variables are the wumpus positions $X_0, \dots, X_T$, each a discrete RV over the G cells under a position encoding $i = \text{row} \cdot W + \text{col}$. The observed variables are the detections $D_t^{(i)}$ for each cell i and time t . The conditional independence assumptions encoded by the model are:

- **Markov dynamics:** $X_{t+1} \perp\!\!\!\perp X_{0:t-1} \mid X_t$.
- **Conditionally independent detections:** given X_t , the per-cell detections at time t are mutually independent of each other and of all detections at other times.

The resulting Bayes net is shown in Figure 1. The joint distribution factorises as

$$p(X_{0:T}, \{D_t^{(i)}\}) = p(X_0) \prod_{t=0}^{T-1} p(X_{t+1} \mid X_t) \prod_{t=0}^T \prod_{i=0}^{G-1} p(D_t^{(i)} \mid X_t). \quad (1)$$

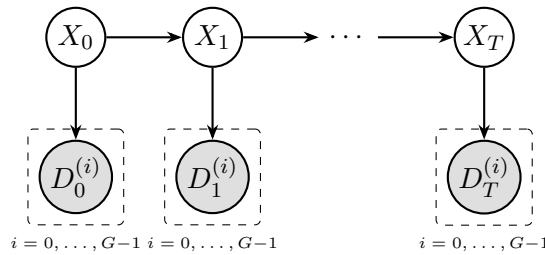


Figure 1: Bayes network for the wumpus tracking model. Each unobserved X_t has G binary detection children.

2.3 Initial-state prior

In the absence of information about the wumpus's starting cell, a uniform prior is used:

$$p(X_0 = x) = \frac{1}{G}, \quad \forall x \in \mathcal{X}. \quad (2)$$

2.4 Transition factor

At each step the wumpus selects a direction $d \in \{U, D, L, R\}$ uniformly and attempts a move. If the target cell is out of grid bounds the move fails and the wumpus remains in place. Marginalising out the unobserved direction, the transition factor is given by:

$$p(X_{t+1} = j | X_t = i) = \frac{1}{4} \sum_{d \in \{U, D, L, R\}} \mathbb{I}[\text{move}(i, d) = j]$$

where $\text{move}(i, d)$ returns the target cell of move d from i , clipped back to i if the target is out of bounds.

The factor is sparse: each row has at most five non-zero entries (at most four in-bounds neighbours plus the cell's own mass carried back from clipped moves). The factor is time-homogeneous, so a single sparse table is built at the beginning and used for every transition.

2.5 Observation factor collapse

Because every detection $D_t^{(i)}$ is observed, the $(T+1)G$ emission factors can be folded into one unary factor per time step on X_t , removing the detection RVs from the graph entirely.

By the conditional independence of detections given the wumpus position (Section 2.2), the joint emission likelihood at time t factorises as:

$$p(\mathbf{d}_t | X_t = x^*) = \prod_{i=0}^{G-1} p(D_t^{(i)} = d_t^{(i)} | X_t = x^*),$$

and each per-cell likelihood depends only on whether cell i contains the wumpus:

$$p(D_t^{(i)} = d_t^{(i)} | X_t = x^*) = \begin{cases} p_w^{d_t^{(i)}} (1-p_w)^{1-d_t^{(i)}} & i = x^*, \\ p_c^{d_t^{(i)}} (1-p_c)^{1-d_t^{(i)}} & i \neq x^*. \end{cases}$$

Next, we can factor out the $i \neq x^*$ terms, which do not depend on x^* and so:

$$p(\mathbf{d}_t | X_t = x^*) = p_w^{d_t^{(x^*)}} (1-p_w)^{1-d_t^{(x^*)}} \cdot \overbrace{\prod_{i \neq x^*} p_c^{d_t^{(i)}} (1-p_c)^{1-d_t^{(i)}}}^{(1)}$$

Where (1) is expanded to the following to enable removal of terms independent of x^* :

$$\frac{\prod_{i=0}^{G-1} p_c^{d_t^{(i)}} (1-p_c)^{1-d_t^{(i)}}}{p_c^{d_t^{(x^*)}} (1-p_c)^{1-d_t^{(x^*)}}}$$

Now:

$$p(\mathbf{d}_t | X_t = x^*) = \underbrace{\left[\prod_{i=0}^{G-1} p_c^{d_t^{(i)}} (1-p_c)^{1-d_t^{(i)}} \right]}_{C_t, \text{ independent of } x^*} \cdot \frac{p_w^{d_t^{(x^*)}} (1-p_w)^{1-d_t^{(x^*)}}}{p_c^{d_t^{(x^*)}} (1-p_c)^{1-d_t^{(x^*)}}}.$$

C_t does not depend on x^* and cancels in any posterior over X_t . We define the unary observation factor:

$$\phi_t(X_t = x^*) \propto \begin{cases} p_w/p_c & d_t^{(x^*)} = 1, \\ (1-p_w)/(1-p_c) & d_t^{(x^*)} = 0. \end{cases}$$

This is a length- G table per time step. The $G(T+1)$ binary detection RVs are absent from the graph used for inference, which becomes $T+1$ unobserved variables with one unary evidence factor per step, a factor of G times smaller than the naive representation.

2.6 Cluster graph and inference algorithm

The factor list (one prior, T transition factors, $T+1$ collapsed observation factors) is handed to `emdws`’s `ClusterGraph` constructor with either the `LTRIP`, `BETHE` or `JTREE` construction tag. For the chain-structured factor graph induced by this model, a non-loopy cluster graph is constructed, so sum-product belief propagation is exact and converges after one forward and one backward sweep. `loopyBP_CG` on this tree-structured CG behaves as standard forward–backward.

Sum-product marginal inference was chosen over MAP for two reasons:

- The per-time-step marginal $\gamma_t(x) = p(X_t | \mathbf{d}_{0:T})$ (i.e. the posterior given all the time steps, not just the preceding ones) is the obvious output for visualisation and for any further purpose of this algorithm. For example, any function that would need to know where the wumpus is would also probably like to know how likely the wumpus is to be there. Saying that the wumpus is in square 1 if it’s only slightly less likely to be in square 2 would be misrepresenting vital information for any real world application of hunting wumpi (or wumpuses)
- the marginals $\gamma_t(x)$ are exactly the quantity required by the E-step of the EM algorithm used for dataset 3 (Section 2.7).

2.7 Parameter learning for unknown p_w, p_c (dataset 3)

For dataset 3, the detection parameters are unknown. We treat them as point estimates to be learned by the EM algorithm. This choice keeps the graph chain-structured (so inference remains exact in each E-step). The alternative of placing Beta priors on p_w, p_c and performing full Bayesian inference would have been possible but was not implemented because it would have caused loopy (and thus non-exact) inference due to the parameters being shared parents of all the observation factors.

The E-step rebuilds the observation factors ϕ_t under the current parameters and runs BP to give the smoothed marginals $\gamma_t^{(k)}(x) = p(X_t = x | \mathbf{d}_{0:T}, \theta^{(k)})$. The M-step has a closed-form solution: with $A_t = \sum_x \gamma_t^{(k)}(x) d_t^{(x)}$ (the expected detection at the wumpus cell at time t) and $D_t = \sum_i d_t^{(i)}$ (the total detection count at time t), the updates are

$$p_w^{(k+1)} = \frac{\sum_t A_t}{T+1}, \quad p_c^{(k+1)} = \frac{\sum_t (D_t - A_t)}{(T+1)(G-1)}.$$

The algorithm alternates between E and M until both of the parameters change by less than 10^{-6} per iteration or 50 iterations are reached. Parameter estimates are clamped to $[10^{-6}, 1-10^{-6}]$ to avoid singular observation factors at boundary values. ?? EM is initialised at $(p_w^{(0)}, p_c^{(0)}) = (0.9, 0.1)$;

2.8 Scope of solution

This work addresses datasets 1–3 in depth and does not extend to datasets 4–5. Choosing to focus on these datasets allowed for thorough analysis of datasets 1–3.

3 Implementation

The implementation comprises two main programs and a shared helper library:

- `miniproject.cc`: datasets 1–2 (known parameters)
- `miniproject_q3.cc`: dataset 3 (EM-based parameter learning)
- `miniproject_helpers.cc`: helper files for detection grids, ground truth, marginals, and trajectories

Both main programs share the same skeleton: load detections, build factors, construct cluster graph, run BP, extract per-time-step marginals, write results to disk, but with the EM variant wrapping

factor build, CG construction, BP, and marginal extraction in an outer loop implementing the M-step of Section 2.7. Each program takes the dataset directory, output directory, initial (p_w, p_c) , and an optional cluster-graph type (LTRIP, JTREE or BETHE) on the command line.

3.1 Variable identifiers and shared cell domain

Each X_t is assigned a sequential integer identifier, and all X_t share a single length- G cell domain instantiated once:

```
vector<RVIdType> X(lastT + 1);
for (int t = 0; t ≤ lastT; ++t) X[t] = static_cast<RVIdType>(t);

rcptr<vector<T>> cellDom(new vector<T>(G));
for (int i = 0; i < G; ++i) (*cellDom)[i] = i;
```

3.2 Building the factors

Prior. The uniform prior on X_0 is encoded by giving an empty map with a default probability of $1/G$, so every entry of the length- G table is automatically $1/G$:

```
map<vector<T>, FProb> emptyProbs;
rcptr<Factor> prior = uniqptr<DT>(new DT(
    {X[0]}, {cellDom}, 1.0 / static_cast<double>(G), emptyProbs));
```

Transition factor. The transition table is built by enumerating each source cell and moving $\frac{1}{4}$ of mass per direction into the right cell. Illegal moves are mapped back to the source cell by `clipMove`, which automatically generates the corner/edge self-loop mass:

```
auto clipMove = [W, H, &cellOf](int row, int col, int drow, int dcol) {
    int nr = row + drow, nc = col + dcol;
    if (nr < 0 || nr ≥ H) nr = row;
    if (nc < 0 || nc ≥ W) nc = col;
    return cellOf(nr, nc);
};

map<vector<T>, FProb> transProbs;
for (int i = 0; i < G; ++i) {
    const int r = rowOf(i), c = colOf(i);
    for (int d = 0; d < 4; ++d) {
        const int j = clipMove(r, c, directions[d][0], directions[d][1]);
        transProbs[{T(i), T(j)}].prob += 0.25;
    }
}
```

The sparse map is built once and reused across all T transition factors, using the fact that all the transitions use the same underlying structure.

Observation factor. Each ϕ_t takes one of the two values $\phi_{\text{on}} = p_w/p_c$ or $\phi_{\text{off}} = (1-p_w)/(1-p_c)$ per cell (see section 2.5), determined by the detection grid at time t :

```
const double phiOn = cfg.pw / cfg.pc;
const double phiOff = (1.0 - cfg.pw) / (1.0 - cfg.pc);

for (int t = 0; t ≤ lastT; ++t) {
    map<vector<T>, FProb> phiProbs;
    for (int row = 0; row < H; ++row)
        for (int col = 0; col < W; ++col) {
```

```

    const int j = cellOf(row, col);
    phiProbs[{T(j)}] = (detections[t][row][col] == 1) ? phiOn : phiOff;
}
rcptr<Factor> phi = uniqptr<DT>(new DT(
    {X[t]}, {cellDom}, defProb, phiProbs));
factors.push_back(phi);
}

```

3.3 Cluster graph and BP

The cluster-graph construction algorithm is selectable at runtime via the command line, enabling the LTRIP/JTREE comparison (see Section 4):

```

map<RvidType, AnyType> obsv;
if (cfg.cgType == "JTREE")
    cgPtr = new ClusterGraph(ClusterGraph::JTREE, factors, obsv);
else if (cfg.cgType == "BETHE")
    cgPtr = new ClusterGraph(ClusterGraph::BETHE, factors, obsv);
else
    cgPtr = new ClusterGraph(ClusterGraph::LTRIP, factors, obsv);

map<Idx2, rcptr<Factor>> msgs; MessageQueue msgQ;
unsigned nMsgs = loopyBP_CG(*cgPtr, msgs, msgQ);

```

3.4 Extracting marginals and the EM loop

For each t the marginal γ_t is obtained by querying the cluster graph for the posterior over X_t and reducing for each cell value:

```

rcptr<Factor> q = queryLBP_CG(cg, msgs, {X[t]})->normalize();
for (int j = 0; j < G; ++j) {
    rcptr<Factor> reduced = q->observeAndReduce({X[t]}, {T(j)});
    gammas[t][j] = reduced->potentialAt(
        emdw::RVids{}, emdw::RVVals{}, true);
}

```

In `miniproject_q3.cc` the E-step code is wrapped in an EM loop. After each E-step the A_t values are computed directly from `gammas[t][j]` and the detection grid, and the M-step updates (p_w, p_c) per the equation in Section (2.7):

```

double sumA = 0.0, sumDminusA = 0.0;
for (int t = 0; t ≤ lastT; ++t) {
    double At = 0.0;
    for (int row = 0; row < H; ++row)
        for (int col = 0; col < W; ++col)
            At += gammas[t][cellOf(row, col)] * static_cast<double>(detections[t][row][col]);
    sumA += At;
    sumDminusA += static_cast<double>(Dt[t]) - At;
}
double newPw = sumA / static_cast<double>(lastT + 1);
double newPc = sumDminusA / (static_cast<double>(lastT + 1) * static_cast<double>(G - 1));
newPw = clamp(newPw, PARAM_EPS, 1.0 - PARAM_EPS);
newPc = clamp(newPc, PARAM_EPS, 1.0 - PARAM_EPS);

```

3.5 Output

For each time step the smoothed marginal is reshaped from its length- G form into a $W \times H$ matrix of probabilities and written to `marginal_<tt>.txt`; the per-step argmax cell forms a trajectory file in the ground-truth coordinate format $(x, y) = (\text{col}, \text{row})$.

4 Results and analysis

4.1 Evaluation methodology

The inferred trajectory is evaluated against the provided ground truth using:

- **Per-step accuracy:** the fraction of time steps for which $\arg \max_x \gamma_t(x)$ equals the ground-truth cell.
- **Mean l_2 cell distance:** average Euclidean distance (in cells) between the inferred argmax cell and the ground-truth cell across time steps.
- **Top-k accuracy:** the fraction of time steps for which the ground-truth cell lies in the k most probable cells under γ_t , reported for $k = 1, 3, 5$.

Qualitative evaluation uses posterior heatmaps for representative time steps with the ground-truth cell overlaid.

4.2 Dataset 1 (5×5 , $p_w=0.95$, $p_c=0.05$)

Dataset 1 was the simplest, with a small grid, and the argmax output was the same as the ground truth for every timestep. See image 2 for the posterior heatmaps of the first 4 frames, the posterior probability was very certain of all wumpus positions for all timesteps.

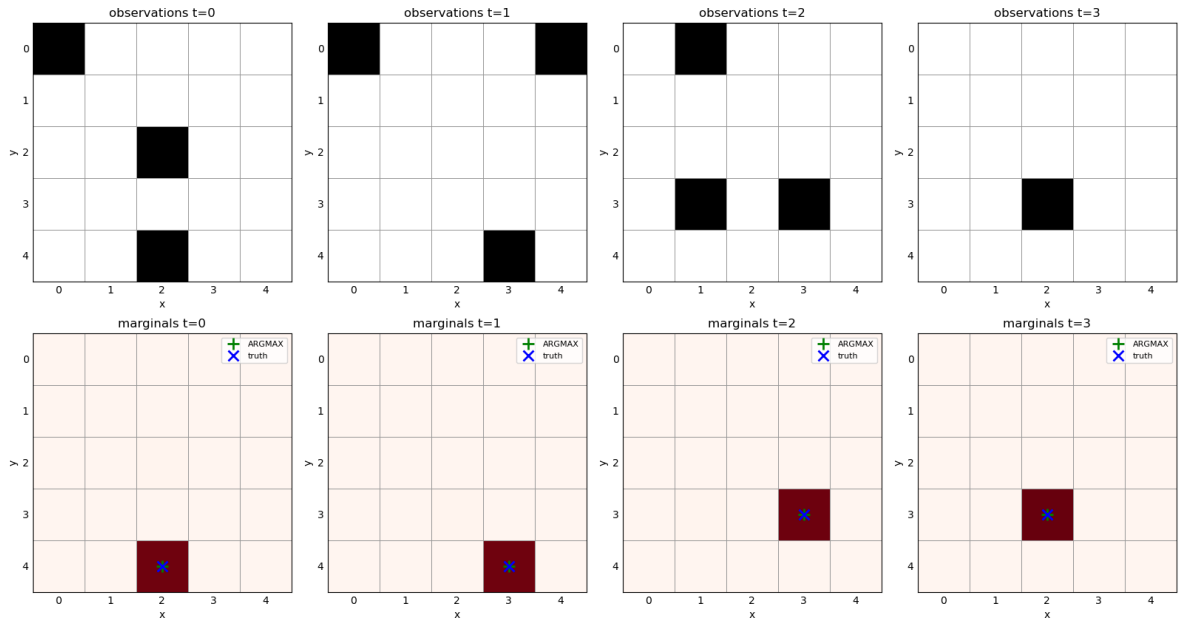


Figure 2: Posterior heatmaps for dataset 1 at the first 4 time steps; ground-truth cell marked with a blue cross, the inferred argmax cell with a green cross and the posterior probability for each cell as the heatmap.

4.3 Dataset 2 (20×20 , $p_w=0.9$, $p_c=0.1$)

The larger grid of dataset 2 had slightly noisier detections, but except for in cases where 2 cells were almost equally likely, the argmax was still correct. For example, in the first frame there was

not enough information to know whether the wumpus was in cell $[15, 16]$ ($p = 0.493$) or $[16, 17]$ ($p = 0.487$) See figure 3 for the posterior heatmaps of the first 4 frames.

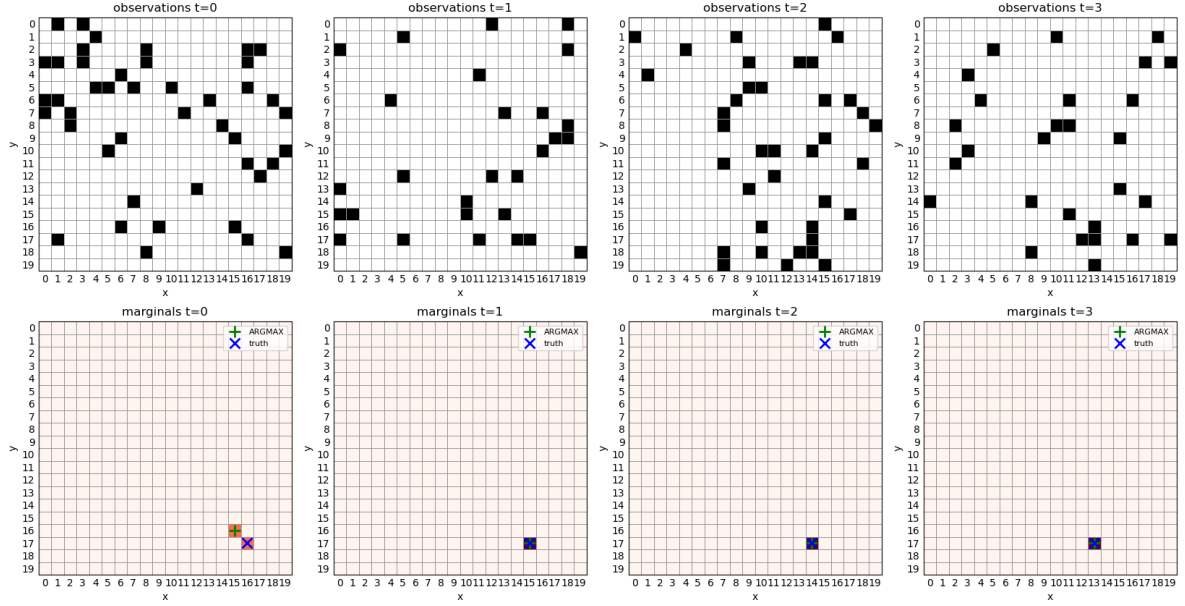


Figure 3: Posterior heatmaps for dataset 2 at the first 4 time steps; ground-truth cell marked with a blue cross, the inferred argmax cell with a green cross and the posterior probability for each cell as the heatmap.

4.4 Dataset 3 (10×20 , EM-learned parameters)

The accuracy on dataset 3 was the worst, however the initial observation frames were highly uninformative due to the noisy observations being in places that the wumpus could conceivably be, and therefore the certainty in the first 4 frames was very low and inconsistent with the ground truth. See figure 4 for the posterior heatmaps of the first 6 frames. After frame 4, the observations were more informative and the posterior certainty increased, however the argmax was still incorrect for time step 6, where the wumpus could have moved to 2 different positions with almost equal probability.

Dataset	Per-step accuracy	Mean cell distance	Top-3 acc.	Top-5 acc.
1	100.0%	0.00	100.0%	100.0%
2	85.0%	0.21	100.0%	100.0%
3	80.0%	0.20	90.0%	100.0%

Table 1: Accuracy results across the three datasets.

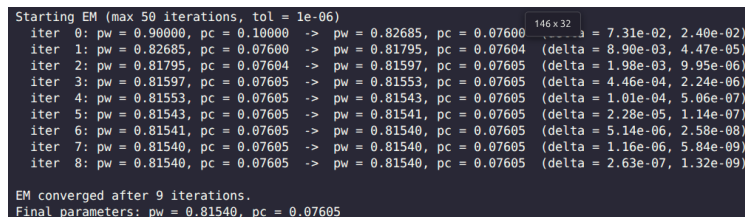


Figure 5: Convergence of p_w and p_c on dataset 3.

EM converged after 9 iterations to $(\hat{p}_w, \hat{p}_c) = (0.815, 0.076)$ with an epsilon of 10^{-6}

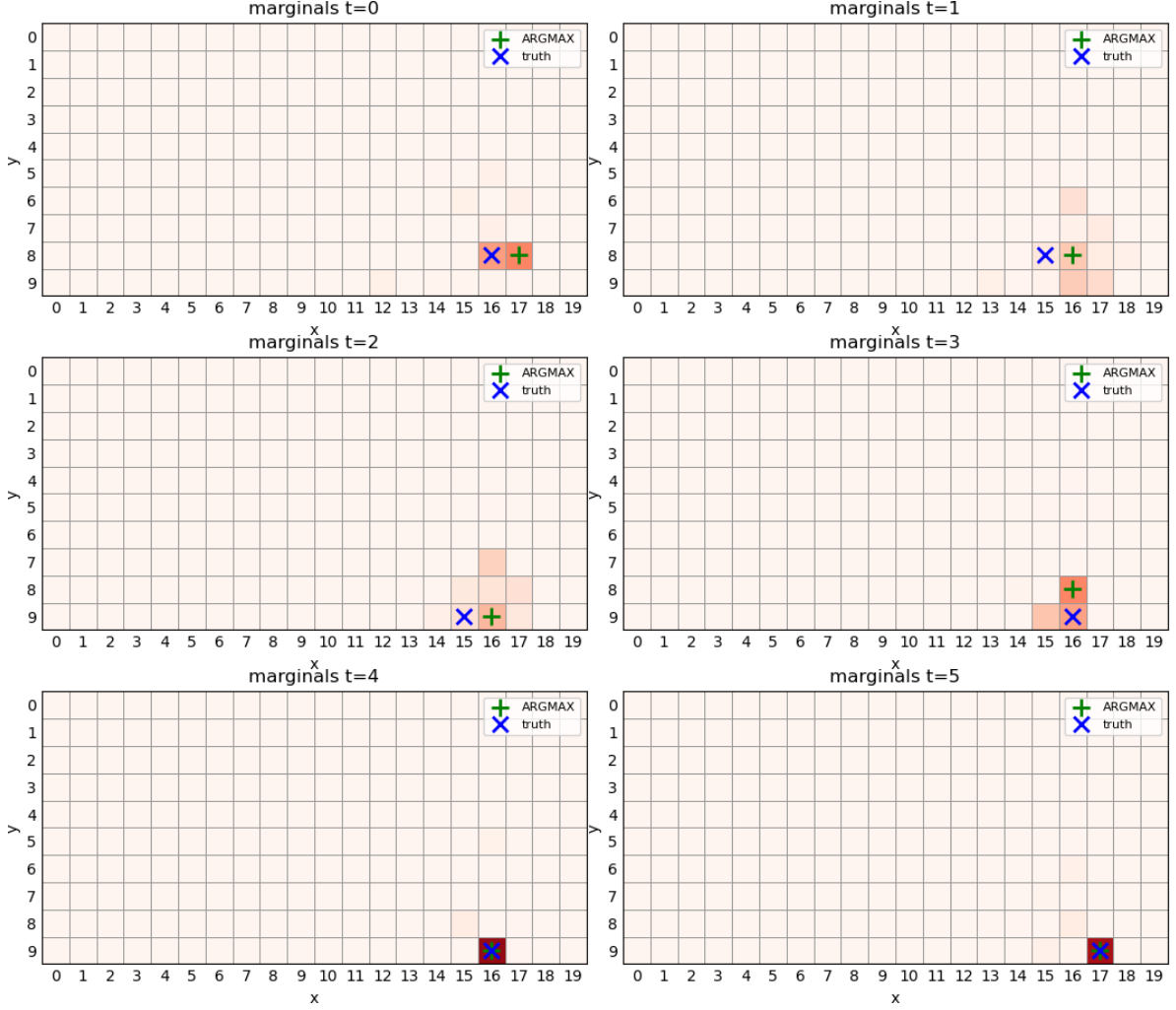


Figure 4: Posterior heatmaps for dataset 3 at the first 6 time steps; ground-truth cell marked with a blue cross, the inferred argmax cell with a green cross and the posterior probability for each cell as the heatmap.

4.5 Cluster-graph choice

Comparing the LTRIP, BETHE and JTREE inference on dataset 3, over many iterations, the mean time for LTRIP was lowest at 0.5s, with JTREE at 0.55s and BETHE at 0.65, all with negligible difference in time variance. In a situation where inference about the current position of the wumpus is necessary as fast as possible, and there is a significant amount of data, LTRIP would be the best choice. If the wumpus is doing what it was originally designed for by Gregory Yob (being hunted), faster inference of the wumpus would be more exact, because the wumpus is running away, and thus changing it's position, as it's position is being inferred.

5 Conclusion

[FILL IN: brief summary paragraph. Suggested points to include: scope (datasets 1–3); key design decisions (HMM Bayes net structure, factor-graph collapse of detection RVs into per-step unary factors, point-estimate EM for unknown parameters, exact sum-product BP on a chain cluster graph); brief acknowledgement of the un-tackled extensions to occupied cells in datasets 4–5 with an indication of the natural extension path (loopy BP over binary map RVs vs alternating EM treating the map as a parameter).]

The HMM-Bayes net approach was successful in inferring the wumpus position accurately most

of the time, and where it was incorrect it was incorrect but unsure of it's correctness. The EM algorithm was able to learn the parameters for dataset 3, however the accuracy was worse than datasets 1 and 2, which is a result that would require further investigation. The cluster graph choice did not have a significant effect on the results, however LTRIP was slightly faster than JTREE and BETHE.

References

- [1] D. Barber, *Bayesian Reasoning and Machine Learning*. Cambridge University Press, 2012.
- [2] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Prentice Hall, 2022.
- [3] H. Koen, H. Roodt, and A. De Waal, "An expert-driven causal model of the rhino poaching problem," *Ecological Modelling*, vol. 347, pp. 29–39, 2017.
- [4] Y. Luo, "Basics of the EM algorithm," Scribe notes for COS 424: Interacting with Data (Lecturer: D. Blei), Lecture 11, Princeton University, Mar. 11, 2008. [Online]. Available: https://www.cs.princeton.edu/courses/archive/spr08/cos424/scribe_notes/0311.pdf